

# CPTG: Structural-Collapse Interpretation of NGC 6946-BH1

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## Abstract

NGC 6946-BH1, also listed as N6946-BH1, is a vanishing-star and failed-supernova candidate whose observed record combines a luminous massive-star progenitor, a weak optical transient in 2009, subsequent optical disappearance, persistent red mid-infrared emission, continued non-recovery in optical monitoring, and no detected Chandra X-ray counterpart. Recent JWST reanalysis separates nearby near-infrared stellar neighbors from the mid-infrared source at the candidate position and models the remnant as a dust-obscured source with luminosity near  $10^{4.7} L_{\odot}$ . This article presents a Curvature Polarization Transport Gravity, or CPTG, structural-collapse interpretation of the same system. In CPTG, compact-object formation is governed by a finite-core structural condition involving physical core mass and curvature-gradient organization, while the external gravitational appearance may be amplified by curvature polarization and organized by curvature transport. Applied to NGC 6946-BH1, the adopted apparent compact-remnant branch  $M_{\text{apparent}} = 22\text{--}27 M_{\odot}$  is compared with a representative CPTG finite-core branch  $M_{\text{core}} = 2\text{--}3 M_{\odot}$ . This gives  $A_{\text{pol}} = 7.33\text{--}13.50$  and a geometric apparent-mass fraction  $f_{\text{geom}} = 0.864\text{--}0.926$ . The same object has a primary remnant-to-progenitor luminosity ratio  $R_L \simeq 0.1245$ , red mid-infrared dust dominance, weak-ejecta scale, optical non-recovery, and X-ray quietness. The result is a mass-energy-SED closure test: NGC 6946-BH1 remains a CPTG finite-core vanishing-star candidate under the adopted observational and mass-branch uncertainties.

## 1 Motivation

NGC 6946-BH1 is one of the best-known candidates for quiet massive-star collapse. The source was monitored before its optical disappearance, brightened in a weak 2009 transient, and later became much fainter than its progenitor in the optical. Historical failed-supernova monitoring identifies the object as a massive-star disappearance candidate rather than a normal luminous core-collapse supernova. Later observations continue to show a red infrared remnant while the optical source remains absent and no Chandra X-ray counterpart is detected [1, 2].

The recent JWST reanalysis makes the object especially useful as a structural-collapse test. Earlier interpretations were complicated by blending: near-infrared

light close to the candidate could be confused with the remnant itself. The newer analysis separates four neighboring near-infrared stellar sources and attributes the relevant mid-infrared emission to NGC 6946-BH1. The remnant is then modeled as a red, dust-obscured source with luminosity around  $10^{4.7} L_{\odot}$ , negligible emission below roughly  $2 \mu\text{m}$ , and a silicate dust shell [4, 6].

CPTG approaches the object from a different physical starting point than the question of whether the source formed a conventional exposed black hole. The CPTG question is whether the observed event is consistent with structural horizon closure: a transition in which a finite compact core crosses a curvature-gradient threshold while its exterior gravitational appearance is amplified by curvature polarization and its remnant environment is organized by curvature transport.

The key observables for this article are therefore not only the progenitor mass or the apparent remnant mass. They are the combined mass-energy-SED closure vector: luminosity suppression, red mid-infrared dust dominance, optical disappearance, X-ray quietness, weak-ejecta scale, and apparent-vs-core mass separation.

## 2 CPTG Structural-Collapse Framework

CPTG is a nonlinear geometric framework in which baryonic matter sources curvature, curvature develops a polarization response, and organized curvature can be redistributed through transport [8]. In the compact-object reduction, the relevant state is not defined by mass alone [9]. It is defined by the physical core mass together with the local curvature-gradient state,

$$(M_{\text{core}}, \chi_{\text{grad}}). \quad (1)$$

The reduced curvature invariant is written as

$$C = g_j g_j + \kappa^2 (\partial_i g_j) (\partial_i g_j), \quad (2)$$

where the first term represents local field strength and the second term represents spatial gradient organization. The dimensionless curvature-response variable is

$$\chi = \frac{C}{a_{\star}^2}, \quad (3)$$

with gradient contribution

$$\chi_{\text{grad}} = \frac{\kappa^2 (\partial_i g_j) (\partial_i g_j)}{a_{\star}^2}. \quad (4)$$

The structural horizon condition is written as

$$\chi_{\text{grad}} \geq \chi_{\text{crit}}. \quad (5)$$

This condition states that compact-object formation is not controlled by enclosed mass alone. A finite core can enter a horizon-like compact state if the curvature-gradient organization reaches the critical threshold. Conversely, mass without the required gradient organization is not sufficient by itself.

The finite-core interior is represented by a regular potential expansion,

$$\Phi(r) = \Phi_0 + \alpha_{\text{core}} r^2 + O(r^4), \quad (6)$$

which gives

$$g(r) = -\nabla\Phi(r) \simeq -2\alpha_{\text{core}} r, \quad (7)$$

and therefore

$$\lim_{r \rightarrow 0} g(r) = 0. \quad (8)$$

The compact interior is finite rather than singular in this reduced description.

The exterior gravitational appearance is not forced to equal the physical baryonic core mass. CPTG writes the apparent-mass relation as

$$M_{\text{apparent}} = A_{\text{pol}} M_{\text{core}}, \quad (9)$$

where  $A_{\text{pol}}$  is the apparent-mass amplification supplied by exterior curvature polarization. Curvature transport does not add matter. It organizes the surrounding remnant geometry, dust, envelope, and fallback environment.

This article therefore treats NGC 6946-BH1 as a mass-energy-SED closure test. The calculation asks whether the observed remnant luminosity, infrared color, optical disappearance, X-ray quietness, weak ejecta, and mass split remain coherent when the apparent compact-remnant mass is separated from the physical finite-core mass.

### 3 Observed Inputs

The observational anchor values used here are the quantities most relevant to the structural-collapse interpretation, drawn from the optical disappearance record, continued monitoring, progenitor-age analysis, and JWST infrared analyses [1, 2, 3, 4, 5, 6]. They are separated into direct observables, upper limits, and adopted CPTG comparison branches.

| Observed quantity             | Adopted value                                                         | Role in CPTG interpretation                   |
|-------------------------------|-----------------------------------------------------------------------|-----------------------------------------------|
| Object                        | NGC 6946-BH1 / N6946-BH1                                              | Failed-supernova and vanishing-star candidate |
| Host galaxy                   | NGC 6946                                                              | Nearby resolved stellar-collapse target       |
| Progenitor scale              | massive red-supergiant branch, historically near $25M_{\odot}$        | Initial stellar-scale context                 |
| Weak 2009 outburst            | peak luminosity above $10^6 L_{\odot}$                                | Low-energy collapse/outburst context          |
| Primary remnant luminosity    | $\log_{10}(L_{\text{rem}}/L_{\odot}) = 4.727$                         | Late remnant energy scale                     |
| Progenitor luminosity branch  | $\log_{10}(L_{\text{prog}}/L_{\odot}) = 5.632$                        | Normalization for luminosity suppression      |
| Luminosity ratio              | $R_L = L_{\text{rem}}/L_{\text{prog}} \simeq 0.1245$                  | Vanishing-star closure scalar                 |
| Mid-infrared behavior         | red MIRI source; silicate dust shell                                  | Dust-remnant and transport environment        |
| Blue/near-infrared behavior   | negligible emission below roughly $2\mu\text{m}$ in the remnant model | Suppression of ordinary exposed stellar SED   |
| Optical recovery              | none detected; upper-limit branch                                     | Survivor/merger stress test                   |
| X-ray counterpart             | none detected by Chandra                                              | Accretion/fallback quietness constraint       |
| Weak ejecta                   | $E_{\text{kin}} \sim 10^{45}\text{--}10^{46}$ erg branch              | Low-energy ejection consistency               |
| Apparent compact-remnant mass | $22\text{--}27M_{\odot}$                                              | Adopted apparent gravitational mass branch    |
| CPTG finite-core mass         | $2\text{--}3M_{\odot}$                                                | Representative physical compact-core branch   |

Table 1: Observed and adopted input branches used for the CPTG structural-collapse interpretation of NGC 6946-BH1.

The central observational sequence is

$$\begin{aligned} \text{weak transient} &\rightarrow \text{optical disappearance} \\ &\rightarrow \text{red mid-infrared remnant} \rightarrow \text{X-ray quietness.} \end{aligned} \quad (10)$$

This sequence does not by itself prove CPTG compact-object physics. It identifies NGC 6946-BH1 as a high-value target for testing whether apparent remnant mass, physical core mass, luminosity suppression, and remnant geometry separate in the way CPTG predicts.

## 4 Luminosity Suppression and Infrared Remnant Scale

The primary luminosity diagnostic is the remnant-to-progenitor ratio

$$R_L = \frac{L_{\text{rem}}}{L_{\text{prog}}}. \quad (11)$$

Using the primary branch [6],

$$\log_{10}(L_{\text{rem}}/L_{\odot}) = 4.727, \quad (12)$$

and

$$\log_{10}(L_{\text{prog}}/L_{\odot}) = 5.632, \quad (13)$$

gives

$$\log_{10} R_L = -0.905, \quad (14)$$

and therefore

$$R_L = 0.1245. \quad (15)$$

With the adopted statistical luminosity uncertainty, the working value is

$$R_L \simeq 0.1245 \pm 0.005. \quad (16)$$

This value is far below unity. It places NGC 6946-BH1 in the dim-remnant branch rather than in the merger-remnant branch where the remnant is expected to be much brighter than the progenitor [6, 7]. The red mid-infrared source, optical disappearance, and low remnant-to-progenitor luminosity are therefore treated as the first closure layer for the CPTG structural-collapse test.

## 5 Mass Split and Apparent-Mass Amplification

The adopted apparent compact-remnant branch is

$$M_{\text{apparent}} = 22\text{--}27M_{\odot}. \quad (17)$$

This is used as an apparent-mass stress branch anchored to the massive-progenitor context rather than as a direct dynamical remnant-mass measurement [1, 3].

For uncertainty propagation this is written as

$$M_{\text{apparent}} = 24.5 \pm 2.5M_{\odot}. \quad (18)$$

The representative CPTG finite-core branch is

$$M_{\text{core}} = 2\text{--}3M_{\odot}, \quad (19)$$

or

$$M_{\text{core}} = 2.5 \pm 0.5M_{\odot}. \quad (20)$$

The apparent-mass amplification is

$$A_{\text{pol}} = \frac{M_{\text{apparent}}}{M_{\text{core}}}. \quad (21)$$

The central value is

$$A_{\text{pol}} = \frac{24.5}{2.5} = 9.80. \quad (22)$$

The propagated fractional uncertainty is

$$\frac{\sigma_A}{A} = \left[ \left( \frac{2.5}{24.5} \right)^2 + \left( \frac{0.5}{2.5} \right)^2 \right]^{1/2} = 0.2245, \quad (23)$$

so that

$$A_{\text{pol}} = 9.80 \pm 2.20. \quad (24)$$

The hard-bound interval is

$$A_{\text{pol}} = 7.33\text{--}13.50. \quad (25)$$

This interval is far from the no-amplification value  $A_{\text{pol}} = 1$ . The apparent-mass variation changes the strength of the inferred curvature-polarization amplification, but it does not change the vanishing-star classification.

## 6 Geometric Apparent-Mass Fraction

The apparent mass can be decomposed into physical finite-core mass plus geometric apparent-mass excess,

$$M_{\text{geom}} = M_{\text{apparent}} - M_{\text{core}}. \quad (26)$$

Using central values,

$$M_{\text{geom}} = 24.5M_{\odot} - 2.5M_{\odot} = 22.0M_{\odot}. \quad (27)$$

With independent branch uncertainties,

$$\sigma_{M_{\text{geom}}} = \sqrt{2.5^2 + 0.5^2} = 2.55M_{\odot}, \quad (28)$$

and therefore

$$M_{\text{geom}} = 22.0 \pm 2.55M_{\odot}. \quad (29)$$

The hard-bound interval is

$$M_{\text{geom}} = 19\text{--}25M_{\odot}. \quad (30)$$

The fractional geometric component is

$$f_{\text{geom}} = 1 - \frac{M_{\text{core}}}{M_{\text{apparent}}}. \quad (31)$$

The central value is

$$f_{\text{geom}} = 1 - \frac{2.5}{24.5} = 0.898. \quad (32)$$

The propagated uncertainty gives

$$f_{\text{geom}} = 0.898 \pm 0.023, \quad (33)$$

with hard-bound interval

$$f_{\text{geom}} = 0.864\text{--}0.926. \quad (34)$$

In this branch, roughly 86–93% of the apparent compact-remnant mass is geometric apparent-mass excess rather than physical finite-core mass. This is the core mass-split result of the CPTG interpretation.

| Quantity              | Central value             | Hard-bound interval       |
|-----------------------|---------------------------|---------------------------|
| $M_{\text{apparent}}$ | $24.5 \pm 2.5 M_{\odot}$  | $22\text{--}27 M_{\odot}$ |
| $M_{\text{core}}$     | $2.5 \pm 0.5 M_{\odot}$   | $2\text{--}3 M_{\odot}$   |
| $A_{\text{pol}}$      | $9.80 \pm 2.20$           | $7.33\text{--}13.50$      |
| $M_{\text{geom}}$     | $22.0 \pm 2.55 M_{\odot}$ | $19\text{--}25 M_{\odot}$ |
| $f_{\text{geom}}$     | $0.898 \pm 0.023$         | $0.864\text{--}0.926$     |

Table 2: CPTG mass-split extraction for the adopted NGC 6946-BH1 apparent-mass and finite-core branches.

## 7 Weak-Ejecta and Quietness Closure

A normal luminous core-collapse supernova ejects a large fraction of the stellar envelope with high kinetic energy. NGC 6946-BH1 instead shows a weak-transient and dim-remnant pattern [1, 6]. The working ejecta branch is

$$M_{\text{ej}} \sim 0.1\text{--}1 M_{\odot}, \quad (35)$$

with representative ejecta speed

$$v_{\text{ej}} \sim 60 \text{ km s}^{-1}. \quad (36)$$

The kinetic scale is therefore

$$E_{\text{kin}} \sim \frac{1}{2} M_{\text{ej}} v_{\text{ej}}^2. \quad (37)$$

For the lower ejecta branch,

$$E_{\text{kin}} \sim 3.6 \times 10^{45} \text{ erg}, \quad (38)$$

and for the upper branch,

$$E_{\text{kin}} \sim 3.6 \times 10^{46} \text{ erg}. \quad (39)$$

This is a weak-ejecta scale, far below the kinetic scale of an ordinary energetic supernova.

The quietness constraints are equally important. The optical source has not reappeared as a normal luminous survivor. The X-ray counterpart is also absent [2]. Using the apparent-mass branch, the X-ray upper-limit ratio is

$$\frac{L_X}{L_{\text{Edd,app}}} \lesssim (1.7\text{--}2.1) \times 10^{-3}. \quad (40)$$

Using the finite-core branch instead gives

$$\frac{L_X}{L_{\text{Edd,core}}} \lesssim 0.015\text{--}0.023. \quad (41)$$

Thus the object remains X-ray quiet under both the apparent gravitational mass and the physical finite-core mass branches.

In the CPTG interpretation, this quietness is not a secondary detail. It is part of the closure condition. A finite core with strong exterior curvature-polarization amplification can appear gravitationally massive without requiring an exposed, luminous, accreting singularity. The red dust shell and weak ejecta then become remnant-geometry observables rather than evidence for a bright recovered star.

## 8 Systematic-Branch Robustness

The primary branch gives [6]

$$R_L = 0.1245 \pm 0.005. \quad (42)$$

A wider systematic branch from the JWST/MIRI and continued-monitoring literature allows [4, 5, 7]

$$R_L = 0.13\text{--}0.25, \quad (43)$$

while a lower mid-infrared branch gives

$$R_L = 0.10\text{--}0.15. \quad (44)$$

All of these remain far below unity. They also remain far from the merger-remnant regime in which the remnant is expected to be brighter than the progenitor.



The Monte Carlo ledger used conservative pass criteria:

$$R_L < 0.5, \quad (45)$$

$$\frac{F_{2100}}{F_{560}} > 1, \quad (46)$$

$$\frac{F_{360}}{F_{560}} < 0.2, \quad (47)$$

$$\frac{L_X}{L_{\text{Edd,app}}} < 10^{-2}, \quad (48)$$

and

$$A_{\text{pol}} > 1. \quad (49)$$

Under the primary branch, the systematic 13–25% luminosity branch, and the lower 10–15% branch, the pass fraction was

$$P_{\text{pass}} = 1.000000. \quad (50)$$

No branch tested moved NGC 6946-BH1 out of the vanishing-star structural-collapse region.

## 9 Interpretive Comparison

The same observational record can be organized in different ways. A conventional failed-supernova interpretation treats the source as a massive star that formed a compact remnant without producing a normal luminous supernova. A dusty-survivor or merger interpretation requires the late-time infrared source to be connected to a surviving or recovering stellar object. The CPTG structural-collapse interpretation instead separates physical finite-core mass from apparent gravitational mass and tests whether the remnant environment remains quiet, dim, red, and weakly ejective.

| Interpretation                       | Expected signature                                                                        | NGC 6946-BH1 comparison                                                         |
|--------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Ordinary luminous supernova          | high kinetic energy, bright optical transient, large ejecta energy                        | Not observed; event was weak and source vanished optically                      |
| Recovered survivor or merger remnant | remnant luminosity comparable to or brighter than progenitor; stellar SED may reappear    | Disfavored by $R_L \simeq 0.1245$ , red dust dominance, and no optical recovery |
| Conventional failed supernova        | weak transient, optical disappearance, compact remnant, fallback/dust emission            | Broadly consistent with the observed vanishing-star record                      |
| CPTG structural collapse             | finite core, apparent-mass amplification, red remnant geometry, quiet X-rays, weak ejecta | Closure-stable under primary and systematic branches                            |

Table 3: Interpretive comparison for NGC 6946-BH1. The CPTG interpretation is evaluated as a structural-collapse closure test rather than as a replacement for the observed failed-supernova phenomenology.

## 10 Falsifiability

The CPTG interpretation is falsifiable by future observations, especially continued JWST and optical/X-ray monitoring [2, 7]. The strongest falsifiers are:

1. Recovery of a normal optical or near-infrared stellar photosphere at the candidate position.
2. A remnant luminosity that rises toward or above the progenitor luminosity in a manner characteristic of a surviving merger remnant.
3. Detection of a persistent bright X-ray source requiring ordinary luminous accretion onto the compact object.
4. Loss of the red mid-infrared dust-dominated SED without a corresponding compact-remnant interpretation.
5. A future dynamical or environmental mass estimate incompatible with the adopted apparent-vs-core mass split.

Conversely, continued optical non-recovery, continued X-ray quietness, and gradual fading or cooling of a red mid-infrared dust shell would strengthen the structural-collapse classification.

## 11 Conclusion

NGC 6946-BH1 is a strong CPTG structural-collapse target because it combines a massive-star progenitor, weak optical transient, optical disappearance, red mid-infrared remnant, X-ray quietness, weak-ejecta scale, and an adopted apparent compact-remnant branch much larger than the representative finite-core branch.

The primary luminosity ratio is

$$R_L = 0.1245 \pm 0.005, \quad (51)$$

placing the remnant firmly in the dim vanishing-star branch. The adopted mass split gives

$$M_{\text{apparent}} = 22\text{--}27M_{\odot}, \quad M_{\text{core}} = 2\text{--}3M_{\odot}, \quad (52)$$

with

$$A_{\text{pol}} = 7.33\text{--}13.50, \quad (53)$$

and

$$f_{\text{geom}} = 0.864\text{--}0.926. \quad (54)$$

The central propagated values are

$$A_{\text{pol}} = 9.80 \pm 2.20, \quad f_{\text{geom}} = 0.898 \pm 0.023. \quad (55)$$

Under the primary JWST branch and the systematic luminosity branches tested here, NGC 6946-BH1 remains closure-stable as a CPTG finite-core vanishing-star candidate. The apparent-mass uncertainty changes the strength of the curvature-polarization amplification but does not change the classification, because the classification is controlled by luminosity suppression, red mid-infrared dust dominance, optical non-recovery, X-ray quietness, weak ejecta, and strong apparent-vs-core mass separation.

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